Yunhan Qiao

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Ph.D. Candidate, Computer Science, Oregon State University

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Research Interests

My research sits at the intersection of empirical software engineering, code comprehension, and human–AI collaboration. I study how students and experienced developers work with GenAI coding assistants and agents (e.g., GitHub Copilot, Cursor) in realistic *brownfield* codebases, and whether faster, more correct work also means deeper understanding. I combine controlled experiments, screen-recorded behavioral analysis, prompt coding, EEG, and interviews to inform how computing education teaches and assesses GenAI-assisted development and how future AI coding agents and IDEs should be designed.

Education

Oregon State University — Corvallis, OR, United States
Ph.D. in Computer Science, School of EECS — *Sep 2023 – Jun 2027 (expected)*
Advisor: Dr. Christopher Hundhausen at the SEAL Lab
Research Areas: Empirical Software Engineering, Code Comprehension, Computing Education, Human–AI Interaction

Arizona State University — Tempe, AZ, United States
M.S. in Software Engineering — *Aug 2021 – May 2023*
Advisor: Dr. Robert LiKamWa
Thesis: Networked System for Volumetric Athletic Coaching in Augmented Reality

Guangxi University of Science and Technology — Liuzhou, China
B.E. in Software Engineering — *Sep 2016 – Jun 2020*

Southern Cross University — Gold Coast, Australia
B.E. in Information Technology — *Sep 2017 – Jun 2020*

Publications (Google Scholar Page)

Peer-Reviewed Journal Articles

[J2] Tariq, A., Conrad, P., Hundhausen, C., Chandrasekaran, S., **Qiao, Y.**, & Haque, S. (2026). Using SlackBot-prompted reflections to promote contribution equality in team software development projects. *ACM Transactions on Computing Education.* https://doi.org/10.1145/3857221

[J1] **Qiao, Y.**, Shihab, M. I. H., & Hundhausen, C. (2026). A systematic literature review of the use of GenAI assistants for code comprehension: Implications for computing education research and practice. *ACM Transactions on Computing Education, 26*(2), Article 24, 1–33. https://doi.org/10.1145/3785366
Selected by the TOCE editorial board for presentation at SIGCSE TS 2027.

Peer-Reviewed Conference Proceedings

[C2] Motwani, M., Kulkarni, A., **Qiao, Y.**, Davis, M., & Chen, Z. (2025). LLM-guided differential fuzzing for detecting platform-specific bugs in scientific applications. *2025 IEEE Conference on AI x Software Engineering (AIxSE)*, 59–66. https://doi.org/10.1109/AIxSE64906.2025.00015

[C1] Shihab, M. I. H., Hundhausen, C., Tariq, A., Haque, S., **Qiao, Y.**, & Mulanda, B. W. (2025). The effects of GitHub Copilot on computing students' programming effectiveness, efficiency, and processes in brownfield coding tasks. *Proceedings of the 2025 ACM Conference on International Computing Education Research (ICER '25), 1*, 407–420. https://doi.org/10.1145/3702652.3744219

Under Review

[R2] **Qiao, Y.**, Haque, S., & Hundhausen, C. (2026). Beyond the prompt: Linking what developers ask, do, and understand with coding agents. Under review at *IEEE International Conference on Software Analysis, Evolution and Reengineering (SANER 2027).* https://arxiv.org/abs/2609.33712

[R1] **Qiao, Y.**, Shihab, M. I. H., Haque, S., & Hundhausen, C. (2025). Faster but not wiser: GitHub Copilot decouples programming performance from code comprehension in brownfield tasks. Under review at *ACM Transactions on Computing Education.* https://arxiv.org/abs/2511.02922

Under Submission

[S1] Schutz, O., **Qiao, Y.**, Moussaoui, F. A., Choudhuri, R., & Sarma, A. (2026). Heart and mind in the loop: Why we vibe when we work with AI. Submitted to *International Conference on Cooperative and Human Aspects of Software Engineering (CHASE 2027).*

Teaching Experience

Graduate Teaching Assistant, School of EECS — Oregon State University
Undergraduate software engineering courses — *2024 – current*

Courses: CS 362 Software Engineering II; CS 461 Senior Software Engineering Project; CS 467 Online Capstone Project; CS 492 Mobile Software Development

Professional Experience

Software Engineering Intern, Mocha Software Nanjing, China
Software development *Aug 2020 – Nov 2020*
Developed a conference system for China Mobile Migu Culture & Technology Group with Spring, Spring MVC, and MyBatis, using Ajax to cut page rendering time from 1 s to 0.5 s and dynamic SQL with Redis-backed SSO sessions to reduce search load.

Awards & Scholarships

Oregon State University Scholarly Presentation Award *2026*
Oregon State University EECS Graduate Scholarship *2023 – current*
Guangxi University of Science and Technology Merit Award *2018*

Academic Service

SIGCSE TS 2027, Main Track, Program Committee

SANER 2027, Short Papers and Posters Track, Program Committee

ACM Transactions on Computing Education (TOCE), Reviewer

Skills & Coursework

Research Methods: Qualitative Analysis, Quantitative Analysis, Interviews, Survey Design
Programming: Python, Java, C, JavaScript, HTML, CSS, SQL, Bash
Frameworks & Tools: React, Express, Flask, MongoDB, PostgreSQL, Git, AWS, Docker, LaTeX, Figma; Claude Code, GitHub Copilot, Cursor, OpenAI Codex
Coursework: Machine Learning; NLP with Deep Learning; Software Engineering for ML; Program Analysis; Algorithms & Data Structures; Human–Computer Interaction; Field Studies in SE & HCI; Methods of Data Analysis

Languages

English (Fluent), Mandarin Chinese (Native)